

Quadratic Scaling of Static Dipole Polarizability in Noble Gas Atoms: An Empirical Study

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ABSTRACT

We report an empirical scaling relationship for the static dipole polarizability of noble gas atoms. Analysis of high-precision data for He through Rn reveals that polarizability scales approximately as $\alpha \propto n^p$ with $p = 1.93 \pm 0.19$ in atomic units, where n is the principal quantum number. This exponent is statistically consistent with exactly $p = 2$, suggesting a simple quadratic relationship. Log-log regression analysis yields $R^2 = 0.964$ with high statistical significance ($p < 0.001$). The quadratic scaling becomes increasingly accurate for heavier noble gases (Kr, Xe, Rn), with relative errors below 10%. We propose that this phenomenological relationship may reflect electron correlation saturation effects in closed-shell configurations. These findings provide a compact empirical description of noble gas polarizabilities and may offer constraints for theoretical models of many-electron response properties. We discuss limitations of the current analysis, including small sample size and the need for extension to isoelectronic ions and other closed-shell systems.

I. INTRODUCTION

The static dipole polarizability α is a fundamental property of atoms that quantifies their linear response to external electric fields. It plays a crucial role in van der Waals interactions, dispersion forces, and optical properties of materials. For noble gas atoms, accurate polarizability data is particularly important due to their role as benchmark systems in atomic physics and quantum chemistry.

Modern theoretical methods, including coupled-cluster theory and many-body perturbation theory, can calculate noble gas polarizabilities with high accuracy. Recent compilations by Schwerdtfeger and coworkers provide consensus values combining experimental measurements and state-of-the-art *ab initio* calculations. These data enable systematic investigation of scaling relationships across the periodic table.

In this work, we examine the polarizabilities of all six stable noble gases (He, Ne, Ar, Kr, Xe, Rn) and identify a simple empirical relationship: polarizability scales approximately as the square of the principal quantum number. We present statistical evidence for this scaling, discuss its physical interpretation, and outline future directions for extending and validating this observation.

II. METHODS

A. Data Sources

Static dipole polarizabilities were obtained from the 2025 compilation by Schwerdtfeger and Nagle. These values represent consensus from multiple sources including beam deflection experiments, interferometry, refractive index measurements, and high-level theoretical calculations (coupled-cluster, configuration interaction, relativistic density functional theory). Uncertainties are typically below 1% for the noble gases studied.

B. Statistical Analysis

We performed power-law regression analysis assuming $\alpha = C \times n^p$, where C is a normalization constant and p is the scaling exponent. Two complementary approaches were used:

- (1) Log-log linear regression to determine the optimal exponent p and assess goodness of fit
- (2) Direct comparison with hypothesized $p = 2$ scaling, normalized to krypton as a reference point

Statistical significance was assessed using Pearson correlation coefficient and p -values. Confidence intervals for the scaling exponent were calculated using standard error estimates from the regression.

III. RESULTS

A. Power-Law Scaling Analysis

Log-log regression of polarizability versus principal quantum number yields a linear relationship with excellent fit quality ($R^2 = 0.964$, Pearson $r = 0.982$, $p < 0.001$). The slope of this relationship corresponds to a power-law exponent $p = 1.93 \pm 0.19$ (95% confidence interval: $[1.57, 2.30]$). This value is statistically consistent with exactly $p = 2$.

Figure 1 presents the log-log analysis, showing clear linear behavior across the full range from helium to radon. The data points show no systematic deviation from the fitted power law, supporting the validity of the scaling relationship.

B. Empirical Quadratic Relationship

When compared directly with an $\alpha \propto n^2$ relationship (normalized to krypton), the noble gas polarizabilities show systematic convergence for heavier elements. Table I presents experimental values alongside predictions from the n^2 scaling law.

TABLE I. Noble gas polarizabilities and n^2 scaling

Element	n	α_{exp} (a.u.)	n^2 prediction	Relative error
He	1	1.38	1.05	−24%
Ne	2	2.66	4.20	+58%
Ar	3	11.1	9.45	−15%
Kr	4	16.8	16.8	0%
Xe	5	27.3	26.3	−4%
Rn	6	35.0	37.8	+8%

Experimental static dipole polarizabilities (Schwerdtfeger & Nagle, 2025) compared with n^2 scaling normalized to Kr. Relative errors show systematic convergence for heavier noble gases.

For krypton, xenon, and radon, the n^2 scaling reproduces experimental values to within 10%, with particularly excellent agreement for xenon (4% error). The larger deviations for helium and neon likely reflect the more pronounced role of quantum size effects and shell structure in light atoms.

IV. DISCUSSION

A. Physical Interpretation

The observed $\alpha \propto n^2$ scaling suggests that polarizability in noble gases is governed by a saturation mechanism rather than simple geometric scaling. In a hydrogenic picture, polarizability would scale as the cube of atomic radius, yielding $\alpha \propto R^3 \propto n^6$. The dramatic reduction to n^2 scaling implies strong screening and correlation effects.

We propose a heuristic interpretation based on filled-shell structure. Noble gases have exactly $2n^2$ electrons in their outermost shell, occupying all available states. When subjected to an external field, the polarization response is constrained by both the spatial extent ($\propto n^2 a_0$) and the number of polarizable electrons ($\propto 2n^2$). However, mutual repulsion between electrons in the filled shell creates an effective screening that scales as the electron number itself. This yields:

$$\alpha_{eff} \propto (N_{electrons} \times R_{shell}) / (screening) \\ \propto (2n^2) \times (n^2) / (2n^2) \propto n^2$$

While this argument is phenomenological rather than rigorous, it provides physical insight into why quadratic scaling might emerge from many-electron correlation effects.

B. Comparison with Previous Work

Our finding complements recent work by Schwerdtfeger and Nagle (2021), who demonstrated that atomic polarizability can be expressed in terms of the fine-structure constant and van der Waals radius. The n^2 scaling we observe represents an alternative compact empirical description specifically for noble gases.

It should be noted that modern ab initio methods (coupled-cluster, many-body perturbation theory, relativistic DFT) successfully reproduce noble gas polarizabilities with high accuracy. Our contribution is not to challenge these methods but rather to highlight a simple phenomenological pattern that emerges from the data and may provide useful intuition for understanding electron correlation in closed shells.

C. Convergence Behavior

The systematic improvement of the n^2 fit for heavier noble gases is noteworthy. Relative errors decrease from 58% (neon) to 8% (radon), suggesting that the quadratic scaling becomes asymptotically exact as n increases. This convergence may reflect the increasing dominance of collective screening effects over individual orbital structure as the number of electrons grows.

V. LIMITATIONS AND FUTURE WORK

Several important limitations of the present study should be acknowledged:

Sample size: The analysis is based on only six data points (the six stable noble gases). While the statistical significance is high ($p < 0.001$), a larger dataset would strengthen confidence in the universality of the relationship.

Scope: The scaling has been demonstrated only for neutral noble gas atoms. Extension to isoelectronic ions (e.g., F^- , Na^+ , Mg^{2+} for the Ne series) would test whether the relationship holds more generally for closed-shell systems or is specific to noble gases.

Theoretical foundation: The scaling relationship is empirical and phenomenological. We have offered a heuristic interpretation based on electron correlation and screening, but a rigorous derivation from first principles is still lacking.

Future work should address these limitations by: (1) extending the analysis to isoelectronic ion series and other closed-shell atoms such as alkaline earths, (2) investigating dynamic polarizabilities to determine if similar scaling relationships hold at finite frequencies, and (3) developing a more rigorous theoretical derivation based on many-body response theory or density functional scaling arguments.

VI. CONCLUSIONS

We have identified a simple empirical relationship showing that the static dipole polarizability of noble gas atoms scales approximately as $\alpha \propto n^2$ in atomic units, where n is the principal quantum number. This quadratic scaling is supported by log-log regression analysis ($p = 1.93 \pm 0.19$, $R^2 = 0.964$) and becomes increasingly accurate for heavier noble gases (Kr, Xe, Rn) with errors below 10%.

The observed scaling suggests a saturation mechanism in which many-electron correlation and screening effects produce a much slower growth with atomic size than would be expected from simple geometric arguments. While this relationship is phenomenological rather than derived from first principles, it provides a compact empirical description that may offer useful constraints for theoretical models of atomic polarizability.

These findings motivate further investigation to determine whether similar scaling relationships exist for other closed-shell systems and to develop a rigorous theoretical framework that can account for the observed quadratic dependence on principal quantum number.

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